LESSON 3.3
EQUIVALENT CONVERSIONS: FRACTIONS PERCENTAGES – PART 2

LESSON INTRODUCTION

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

LEARNING TARGETS

- I can convert between fractions and percentages.
- I can rewrite positive rational numbers in different but equivalent forms.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.

WORKING TOWARD

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

ESSENTIAL QUESTIONS

- What strategies can be used to convert fractions and decimals?
- What strategies can be used to convert fractions and percentages?
- Why is it helpful to convert a fraction into a decimal first when converting to percentages?

LESSON NARRATIVE

In this lesson, students continue to convert between fractions and percentages, now focusing on rewriting positive rational numbers in all three different but equivalent rational forms: decimals, fractions, and percentages. The Guided Instruction positions students to develop fluency when rewriting positive rational numbers in different but equivalent rational forms, by recognizing when it is helpful to first convert fractions to decimals if the denominator is not a factor of 100. Students then work in partners to develop and discuss strategies for rewriting rational numbers, and practice computational accuracy.

COMPONENT SUMMARY

3.3.1 WARM-UP *(~5 MIN.)*

Matching decimals with equivalent percentages puzzle

3.3.2 GUIDED INSTRUCTION *(20-25 MIN.)*

Guided review on how to convert between decimals, fractions, and percentages

3.3.3 COLLABORATION *(15-20 MIN.)*

Word Bank for formalizing strategies when converting between fractions and percentages

3.3.4 GUIDED PRACTICE *(10 -15 MIN.)*

Individual practice rewriting rational numbers in equivalent forms

3.3.5 WRAP-UP *(~5 MIN.)*

Assessment of learning targets rewriting rational numbers in equivalent forms

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Percent

MATERIALS

- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- If blank versions of the tables are being used as suggested in Differentiate Instruction, prepare these tables on chart paper prior to the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble remembering to divide the numerator by the denominator when converting fractions to decimals.
- Students may not recognize equivalent fractions.
- Students may confuse when to multiply or divide by 100 when converting between decimals and percentages.

THINGS TO CONSIDER

- Consider emphasizing equivalent fractions and strategies for finding equivalent fractions at the start of the lesson.
- Encourage students to use multiplication and division reasoning as they write equivalent fractions.
- Consider further building students' fluency with factors and the greatest common factor and least common multiple.
- If time is a constraint, assign students 3.3.4 Guided Practice as homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

3.3.2 Guided Instruction provides students with an opportunity to compare converting fractions, decimals, and percentages. Students can discuss the similarities and differences in calculating the equivalent forms in all three tables in this activity. Consider writing student strategies in a shared classroom space for consideration in 3.3.3 Collaboration.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Throughout this lesson, students engage in discussions that reflect on their own mathematical thinking and others' thinking in relation to strategies for rewriting rational numbers in equivalent forms. In 3.3.3 Collaboration, students utilize mathematical vocabulary while working together to analyze and communicate methods needed to complete the incomplete strategy statements.

DIFFERENTIATE INSTRUCTION

As an entry point for kinesthetic learners, blank versions of the tables in 3.3.2 Guided Instruction and word bank activities in 3.3.3 Collaboration can be displayed on chart to provide the opportunity for learners to move around the room while working, or work in jigsaw groups to finish one table and present their learning to another.

Consider using observations of students' understanding from 3.3.2 Guided Instruction to pull students for small-group instruction for 3.3.4 Guided Practice. Consider displaying the five strategy statements from 3.3.3 Collaboration in a shared classroom space to anchor the various strategies for visual learners.

3.3.1 WARM-UP: MATCHING PUZZLE

Component Overview: *In this activity, ask students to individually attempt to complete the puzzle by matching decimals to their equivalent percentages, then check their work with a partner. Then have students share the strategies they used to determine the decimal and percentage equivalences.*

TEACHER MOVES

Consider having students verbally describe how they determined the equivalent percentage. Highlight strategies that use place value and the connection between percentages representing values "per 100."

Questioning to address misconceptions related to converting decimals and percentages might include:

- How does the decimal form you connected to the percentage relate to a value out of 100?
- What do you notice about how the decimal placement in the decimal compares to the percent form? How does this connect to a strategy for converting decimals to percentages?
- Would you divide or multiply by 100 to rewrite decimals as percentages? Why?

3.3.1 Warm-Up: Matching Puzzle

Several values, expressed as decimals and percentages, are given.

0.71	710%
7.1	7.1%
0.071	71%

1. Match the values in the left column with their equivalent form in the right column by drawing lines to connect them.
   See above.

2. Describe how you determined which values are equivalent.
   Answers may vary. I multiplied each value in the left table by 100 and matched it to the corresponding value in the right table.

3.3.2 Guided Instruction: Rewriting Rational Numbers in Equivalent Forms

Recall from Lesson 1 that a fraction can be converted into its equivalent decimal form by dividing the ***numerator*** by the ***denominator***.

1. Complete the table by rewriting the fraction into an equivalent decimal form.

Fraction Form	Decimal Form
$\frac{13}{40}$	0.325
$\frac{7}{8}$	0.875
$2\frac{6}{10}$	2.6

2. Complete the table by rewriting the decimal into an equivalent percentage.

Decimal Form	Percent Form
0.325	32.5%
0.875	87.5%
2.6	260%

3. What is the equivalent decimal form of $\frac{5}{8}$?
   0.625

3.3.2 GUIDED INSTRUCTION: REWRITING RATIONAL NUMBERS IN EQUIVALENT FORMS

Component Overview: *Students work individually or in small groups to convert between decimals, fractions, and percentages. Consider guiding students through the first table in question 1 in this section as a whole group to review how to divide fractions to convert to decimals, then students can work independently or in small groups to complete the problems in the rest of this section. After finishing the tables, arrange students in small groups to discuss methods and strategies used in the conversions.*

TEACHER MOVES

Compare and Connect

After students complete the tables in questions 1 and 2, consider having students share strategies in small groups to compare the processes before sharing whole group.

Consider writing these strategies on a shared classroom space for students to reference throughout the lesson and to connect to the strategy statements in 3.3.3 Collaboration.

ENRICHMENT

To encourage mathematical reasoning as students complete the tables, ask questions like:

- How do you know those forms are equivalent?
- For the table in question 1, how was your strategy different when converting $\frac{13}{40}$ and $\frac{7}{8}$? Did you use equivalent fractions or division? Why?
- Should a number less than one, in fraction form, also have an equivalent decimal form that is less than 1? What about from decimal to percent form?

3.3.2 GUIDED INSTRUCTION: REWRITING RATIONAL NUMBERS IN EQUIVALENT FORMS (CONTINUED)

TEACHER MOVES

Compare and Connect

Consider giving students an opportunity to share with a partner or whole group, targeting benchmark-aligned concepts.

- Connection between powers of 10 and place value, including percentage as "out of 100."
- Purpose of multiplying or dividing by 100, or the connection to "moving" the decimal point.

DIFFERENTIATE INSTRUCTION

Using observations on student understanding in 3.3.2 Guided Instruction, consider using the table in question 7 as an opportunity for a teacher-facilitated small group for learners who need additional support. This small group can take place while other students work on 3.3.3 Collaboration.

4. **What is 0.625 rewritten as a percentage?**
   62.5%

5. **What is $\frac{5}{8}$ rewritten as a percentage?**
   62.5%

6. **When rewriting a fraction as a percentage, why might it be helpful to first rewrite it in its equivalent decimal form?**
   Answers will vary. Rewriting a fraction as its equivalent decimal can make converting the fraction to a percentage easier because the decimal form only requires you to multiply by 100, or move the decimal place two places to the right.

7. **Complete the table by rewriting each value in its equivalent forms.**

Fraction Form	Decimal Form	Percent Form
$\frac{675}{1000} = \frac{27}{40}$	0.675	67.5%
$\frac{19}{1000}$	0.019	1.9%
$\frac{63}{20}$	3.15	315%

3.3.3 Collaboration: Strategies for Rewriting Rational Numbers

Working with your partner, use the words in the bank to complete each statement. Words can be used more than once.

Word Bank

decimal	denominator	fraction
numerator	percentage	place value

1. To convert a *fraction* to a decimal, divide the *numerator* by the *denominator*.
2. Multiply the *decimal* by 100 to convert a *decimal* to a *percentage*.
3. To rewrite a *percentage* as a *decimal*, divide the *percentage* by 100.
4. Use the *place value* of the final digit to determine the denominator of the decimal when rewriting a *decimal* as an equivalent *fraction*.
5. An equivalent fraction with a *denominator* of 100 can be used to convert a *fraction* to a percentage.

3.3.3 COLLABORATION: STRATEGIES FOR REWRITING RATIONAL NUMBERS

Component Overview: *Students use a word bank to complete five statements describing different strategies for rewriting rational numbers in equivalent form. Additionally, consider having students write these statements and corresponding examples in a math journal or notebook to reference in upcoming lessons.*

DIFFERENTIATE INSTRUCTION

Consider opportunities to extend this activity for a variety of learning environments.

- Roles: Each partner has a role; one student only reads the statements and the other matches the correct word (pair strategically).
- Charts: Display each statement on chart paper around the room and students have slips of paper with each word to tape or glue onto each poster.
- Matches: Students are given different words or statements and must walk around the room to find their “match.”

TEACHER MOVES

If students get stuck, consider prompting them to recognize or apply certain examples or strategies they utilized in previous activities to statements they are reading.

3.3.4 GUIDED PRACTICE: REWRITING RATIONAL NUMBERS IN EQUIVALENT FORMS

Component Overview: *Students practice converting between different forms of rational numbers by completing the missing components of a table. The table includes a mix of different forms to assess students' ability to move from any given form to another.*

TEACHER MOVES

Compare and Connect

Consider opportunities for students to connect the strategy they used in these conversions with their statements in the previous activity. This will formalize their learning and processes.

3.3.4 Guided Practice: Rewriting Rational Numbers in Equivalent Forms

1. **Complete the table by rewriting each value in its equivalent forms.**

Fraction Form	Decimal Form	Percent Form
$\frac{3}{8}$	0.375	37.5%
$\frac{22}{100} = \frac{11}{50}$	0.22	22%
$\frac{4}{100} = \frac{1}{25}$	0.04	4%
$2\frac{33}{50}$	2.66	266%

3.3.5 Wrap-Up: Ticket Out the Door

1. **Rewrite each value as an equivalent percentage.**

   a. 0.02 2% b. $\frac{1}{5}$ 20% c. $\frac{11}{8}$ 137.5%

2. **Rewrite each value as an equivalent fraction.**

   a. 67% $\frac{67}{100}$ b. 0.375 $\frac{3}{8}$ c. 35.65 $35\frac{13}{20}$

3.3.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *This Wrap-Up assesses students' abilities to convert between different forms of rational numbers (fractions, decimals, and percentages). Have students work individually as a formative assessment on the learning targets.*

TEACHER MOVES

Students who write 72.7% as their response for $\frac{11}{8}$ in question 1 inverted the dividend and divisor when calculating the percentage.

TEACHER MOVES

Notice the use of the thousandths place in 0.375 for question 2. Identify students who write $\frac{37.5}{100}$ as well as students who write $\frac{375}{1000}$. Have those students compare their responses to determine why they are equivalent, or highlight these responses in the Warm-Up for Lesson 4.